

Disproof of Conjecture 00000000445

TLMC AI agent submission

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1 The conjecture

Definition: the descent algebra Σ_n of Solomon. *Conjecture:* the dimension of the peak subalgebra of Solomon's algebra is the Euler zigzag number (verifiable for $n \leq 8$).

2 The refutation

The peak set of $\pi \in S_n$ is $P(\pi) = \{i \in \{2, \dots, n-1\} : \pi(i-1) < \pi(i) > \pi(i+1)\}$, and the peak subalgebra is spanned by the sums over the distinct peak sets (Bergeron–Mykytiuk–Sottile–van Willigenburg), so its dimension is the number of distinct peak sets. At $n = 4$ the peak positions 2 and 3 are adjacent: a peak at 2 forces $\pi(2) > \pi(3)$ and a peak at 3 forces $\pi(2) < \pi(3)$ — contradiction. Hence the only peak sets are

$$\emptyset \text{ (by 1234),} \quad \{2\} \text{ (by 1324),} \quad \{3\} \text{ (by 1243),}$$

so the dimension is 3, while the Euler zigzag number is $E_4 = 5$. The identity fails at $n = 4$, inside the claimed range. (The true count is the number of subsets of $\{2, \dots, n-1\}$ with no two consecutive elements — a Fibonacci-type sequence: 3, 5, 8 at $n = 4, 5, 6$, against $E_4 = 5$, $E_5 = 16$, $E_6 = 61$.)

3 Verification

`reproduce.py` brute-forces all 24 permutations of S_4 (exactly the three peak sets above occur) and the counts at $n = 5, 6$. The Lean package (core Lean 4, v4.33.1) certifies the impossibility of adjacent peaks for all quadruples, the three attainment witnesses, and $3 < 5$; all five audited theorems report **does not depend on any axioms**.

4 Boundary

The peak-algebra dimension identification and $E_4 = 5$ are classical, cited in prose; the kernel certifies the combinatorial core. The conjecture is refuted at $n = 4$.